mutated F128L
ATP13A2
mutated G504R
ATP13A2
ATP13A2
misfolded
Mn2+
neuron death
cellular manganese ion homeostasis
ATP13A2
protection from oxidative stress
Ni2+
cellular nickel ion homeostasis
Zn2+
cellular zinc ion homeostasis
cellular response to stress
UPS
ERAD pathway
mutated
GBA1
unfolded protein response
ER stress
GBA1
SCARB2
SCARB2
GBA1
LIMP2:GBA1
ER
SCARB2
GBA1
LIMP2:GBA1
golgi
SNARE
SNCA
protofibril
dopaminoquinone
SNCA
L-dopaquinone
SNCA
ER
Ca2+
ITPR1
ITPR2
ITPR3
IP3 receptor
RYR1
RYR2
RYR3
Ryanodine receptor
ATP13A2
LAMP2
LAMP2 multimer
SNCA
protofibril
SNCA
SCARB2
GBA1
LIMP2:GBA1
N-acylsphingosine
alpha-D-glucose
glucosylceramide
mutated G2019S
LRRK2
UPS
chaperone-mediated autophagy
LRRK2
&
macroautophagy
SNCA
Ub
PRKN
CAPN1
CAPNS1
Calpain-1
SCARB2
GBA1
mutated A53T
SNCA
HSPA8
Ca2+
Ca2+
SNCA
SNCA
SNCA
CAPN1
CAPNS1
Calpain-1
mutated A53T
SNCA
LRRK2
TPCN2
PSAP
SNCA
misfolded
SNCA
misfolded
SNCA
misfolded
PLK3
omegasome
NAADP
PLK2
SNCA
S129
P
pSNCA129:PLK2
SNCA
S129
P
omegasome
ATP6V1A
ATP6V1C2
ATP6V1G1
ATP6V1B1
ATP6V1D
ATP6V1G2
ATP6V1B2
ATP6V1E1
ATP6V1H
ATP6V1C1
ATP6V1E2
V1 subunit
ATP6V0A1
ATP6V0C
ATP6V0E2
ATP6V0A2
ATP6V0D1
TCIRG1
ATP6V0D2
ATP6V0B
ATP6V0E1
V0 subunit
V-type proton ATPase
PSEN1
CTSD (32kDa)
CTSD (44kDa)
PLK2
lysosomal lumen acidification
ATP13A2
H+
H+
chaperone-mediated autophagy
Zn2+
Zn2+
MT3
GAPDH
RNASE1
generic substrates of CMA
GAPDH
RNASE1
generic substrates of CMA
GAPDH
RNASE1
generic substrates of CMA
phagophore
phagophore membrane
UBQLN1
UBQLN2
UBQLN4
ubiquilin
LRRK2
LRRK2
LRRK2
mutated G2019S
LRRK2
mutated G2019S
LRRK2
HSPA8
HSPA4
LAMP3
activation of LRRK2 protein kinase activity
phagophore
lysosome
lysosome
LAMP2
LAMP2 multimer
LAMP2
autophagosome
autolysosome
autophagosome
autolysosome
lysosome organization
rapamycin
neuron

ER
ITPR1
ITPR2
ITPR3
IP3 receptor
RYR1
RYR2
RYR3
Ryanodine receptor
SNARE
mutated
GBA1
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ERAD pathway
CAPN1
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Calpain-1
rapamycin
omegasome
UBQLN1
UBQLN2
UBQLN4
ubiquilin
ER
unfolded protein response
phagophore
phagophore
phagophore membrane
lysosome organization
PSEN1
ER stress
PRKN
SNCA
PLK3
PLK2
autophagosome
LRRK2
lysosome
LAMP3
HSPA4
ATP13A2
ATP6V1A
ATP6V1C2
ATP6V1G1
ATP6V1B1
ATP6V1D
ATP6V1G2
ATP6V1B2
ATP6V1E1
ATP6V1H
ATP6V1C1
ATP6V1E2
V1 subunit
ATP6V0A1
ATP6V0C
ATP6V0E2
ATP6V0A2
ATP6V0D1
TCIRG1
ATP6V0D2
ATP6V0B
ATP6V0E1
V0 subunit
V-type proton ATPase
NAADP
Zn2+
Mn2+
SNCA
Ub
SNCA
S129
P
autolysosome
mutated G2019S
LRRK2
LAMP2
Zn2+
H+
TPCN2
MT3
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mutated F128L
ATP13A2
mutated G504R
ATP13A2
PLK2
SNCA
S129
P
pSNCA129:PLK2
macroautophagy
mutated A53T
SNCA
L-dopaquinone
SNCA
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SNCA
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